Proof that $C(\theta)$ is Not Convex

We are given:

$$C(\theta) = \left(\frac{1 - e^{b\theta_0}}{b} - \frac{1 - e^{-a\theta_1}}{a} \right)^2,$$

where $a > 0, b > 0, \theta_0 < 0, \theta_1 > 0$. Define:

$$s(\theta) = \frac{1}{b} - \frac{1}{a} - \frac{e^{b\theta_0}}{b} + \frac{e^{-a\theta_1}}{a}.$$

Thus:

$$C(\theta) = s(\theta)^2.$$

Step 1: Gradient and Hessian of $s(\theta)$

$$\nabla s(\theta) = \begin{bmatrix} \frac{\partial s}{\partial \theta_0} \\ \frac{\partial s}{\partial \theta_1} \end{bmatrix} = \begin{bmatrix} -e^{b\theta_0} \\ -e^{-a\theta_1} \end{bmatrix}.$$

The Hessian of $s(\theta)$ is:

$$\nabla^2 s(\theta) = \begin{bmatrix} -be^{b\theta_0} & 0 \\ 0 & ae^{-a\theta_1} \end{bmatrix}.$$

Step 2: Hessian of $C(\theta)$

For a function $f = s^2$, the Hessian is:

$$\nabla^2 f = 2\nabla s \nabla s^\top + 2s\nabla^2 s.$$

Therefore:

$$H(\theta) = \begin{bmatrix} 2e^{2b\theta_0} - 2bs(\theta)e^{b\theta_0} & 2e^{b\theta_0 - a\theta_1} \\ 2e^{b\theta_0 - a\theta_1} & 2e^{-2a\theta_1} + 2as(\theta)e^{-a\theta_1} \end{bmatrix}.$$

Step 3: Check Positive Semidefinite (PSD) Condition

A function is convex if its Hessian is PSD for all θ. For $H(\theta)$ to be PSD:

$$H_{11} \geq 0, \ H_{22} \geq 0, \ \det(H) \geq 0.$$

Step 4: Counterexample

Take:

$$a = b = 0.5, \quad \theta_0 = -2, \quad \theta_1 = 0.1.$$

Compute:

$$s(\theta) = \frac{1}{0.5} - \frac{1}{0.5} - \frac{e^{-1}}{0.5} + \frac{e^{-0.05}}{0.5} \approx -1.213.$$

Now compute Hessian entries numerically:

$$H_{11} \approx 1.08, \ H_{22} \approx 0.12, \ H_{12} \approx 0.74.$$

Thus:

$$\det(H) = H_{11}H_{22} - H_{12}^2 \approx -0.953 < 0.$$

Since the determinant is negative, the Hessian is indefinite, so $C(\theta)$ is not convex.

Conclusion

$$\boxed{C(\theta) \text{ is not convex in general.}}$$